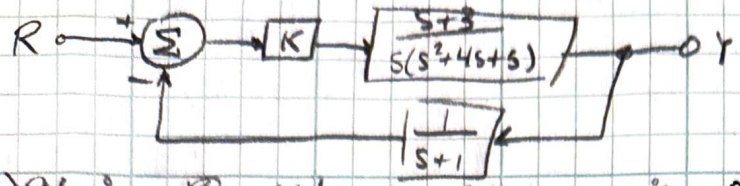
Seth Ricks

ECEN 470
10/26/2026

PSO7: 5.20(a)(b), 5.25(a)(b), 5.27

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5.20



a.) Using Routh's stability criterion, determine all values of k for which the system is stable

b.) Use MATLAB to draw the root locus versus k and find the values of k at the im-axis crossing.

a.) $$H(s) = \frac{\frac{K(s+3)}{s(s^2+4s+5)}}{1+\frac{K(s+3)}{s(s^2+4s+5)(s+1)}} = \frac{\frac{K(s+3)}{s(s^2+4s+5)}}{\frac{s(s^2+4s+5)(s+1)+K(s+3)}{s(s^2+4s+5)(s+1)}}$$

$$= \frac{K(s+3)}{s(s^2+4s+5)} * \frac{s(s^2+4s+5)(s+1)}{s(s^2+4s+5)(s+1)+K(s+3)} = \frac{K(s+3)(s+1)}{s(s^2+4s+5)(s+1)+K(s+3)}$$

Char eq: $s(s^2+4s+5)(s+1)+K(s+3)$

$= (s^3+4s^2+5s)(s+1)+ks+3k$

$= s^4+s^3+4s^3+4s^2+5s^2+5s+ks+3k$

$= s^4+5s^3+9s^2+(5+k)s+3k$

$K(s^2+4s+3)$

$= Ks^2+4ks+3k$

Routh Array:

s^4 :	1	9	3k
s^3 :	5	5+k	0
s^2 :	8−0.2k	3k	0
s^1 :	$\frac{k^2+40k-200}{k+40}$	0	
s^0 :	3k		

$\frac{(5)(9)-1(5+k)}{5} = \frac{45-5-k}{5} = 8-0.2k$

$\frac{(5)(3k)-0}{5} = 3k$

$\frac{(8-0.2k)(5+k)-(5)(3k)}{8-0.2k}$

$= \frac{40-k+8k-0.2k^2-15k}{8-0.2k} * \frac{5}{5}$

$= \frac{200-5k+40k-k^2-75k}{40-k}$

$= \frac{200-40k-k^2}{40-k} * \frac{-1}{-1}$

$= \frac{k^2+40k-200}{k-40}$

$3k>0 \rightarrow k>0$

$8-0.2k>0 \rightarrow -0.2k>-8$

$k<40$

$\frac{k^2+40k-200}{k-40} > 0$

$\frac{(k-(-10\sqrt{6}+20))(k+(10\sqrt{6}+20))}{k-40} > 0$

$\frac{(k-4.49)(k+44.49)}{k-40} > 0$

$k>40$

$-44.49 < k < 4.49$

$\boxed{0<k<4.49}$

b.) Chareq: $1 + k\frac{s+3}{s(s^2+4s+5)(s+1)} = 0$

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$1 + k\frac{s+3}{s^4+5s^3+9s^2+5s} = 0$

$a(s) = s+3$
$b(s) = s^4+5s^3+9s^2+5s$

SEE MATLAB: k where loci crosses imaginary:
k = 0; k = 4.5

↳ 0 is technically not crossing, but it is on the Im axis

problem513.m | problem520a.m | problem51ciii.mlx

```
% Problem 5.20 b
clf;
sys = tf([1 3], [1 5 9 5 0]);
[r, k] = rlocus(sys, 0:0.1:50);

rlocus(sys); grid on; hold on;

for i = 1:length(k)

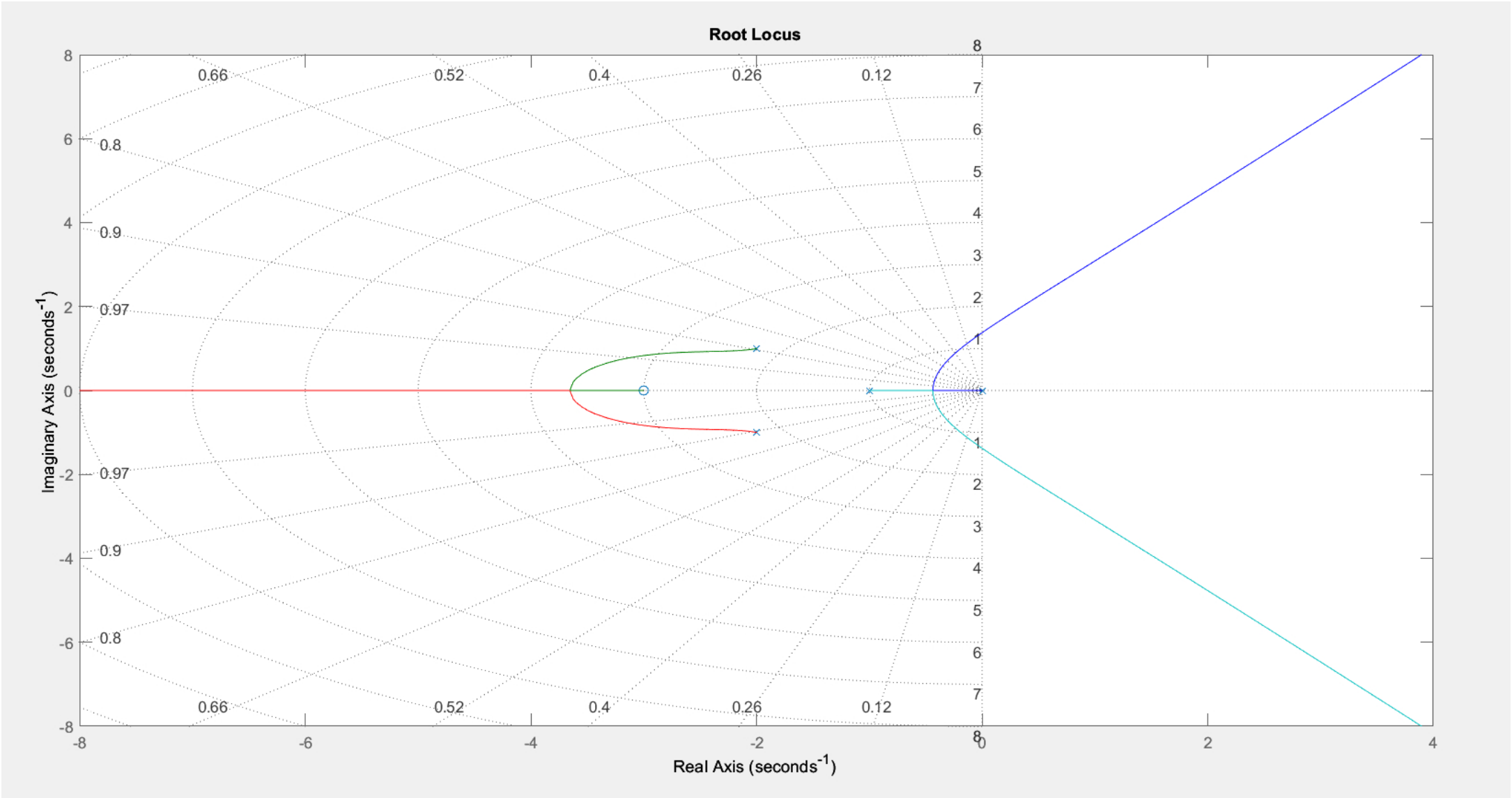
    idx = find(abs(real(r(:, i))) < 1e-3);

    if ~isempty(idx)
        fprintf('Imaginary crossing at K = %.2f | Pole = %.4fi\n', k(
    end
end
hold off;
```

```
>> problem520a
Imaginary crossing at K = 0.00 | Pole = 0.0000i
Imaginary crossing at K = 4.50 | Pole = 1.3787i
>>
```

Workspace

Name	Value
i	501
idx	*[ ]*
k	*1x501 double*
r	*4x501 complex double*
sys	*1x1 tf*


Root Locus
Imaginary Axis (seconds^{-1})
Real Axis (seconds^{-1})
0.66
0.52
0.4
0.26
0.12
0.8
0.9
0.97
8
6
4
2
0
-2
-4
-6
-8
-8
-6
-4
-2
0
2
4
7
5
3
1

(5.25) A numerically controlled machine tool positioning servomechanism has a normalized 1st-type TF: $G(s) = \frac{1}{s(s+1)}$ 3/4

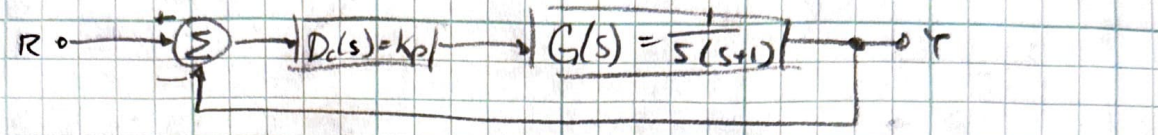
Performance of the system in the unity feedback configuration are satisfied if the closed-loop poles are $s = -1 \pm j\sqrt{3}$

a.) show this specification can't be achieved by proportional alone, $D_c(s) = k_p$

b.) Design lead compensator $D_c(s) = k\frac{s+z}{s+p}$ that will meet this specification

---

a.)

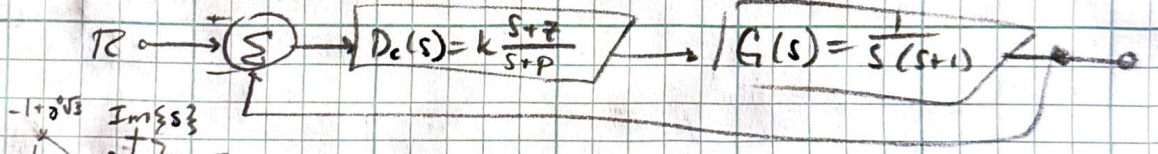


$$H(s) = \frac{\frac{k_p}{s(s+1)}}{1+\frac{k_p}{s(s+1)}} = \frac{\frac{k_p}{s(s+1)}}{\frac{s(s+1)+k_p}{s(s+1)}} = \frac{k_p}{s(s+1)+k_p} = \frac{k_p}{s^2+s+k_p}$$

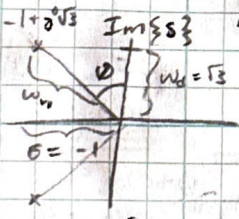
$$s = -1 \pm \frac{\sqrt{1-4k_p}}{2} = -\frac{1}{2} \pm \frac{\sqrt{1-4k_p}}{2}$$

*Not possible b/c real part of poles is $-\frac{1}{2}$

b.) $D_c(s) = k\frac{s+z}{s+p}$, $z < p$



$s = -1 \pm j\sqrt{3}$



$\omega_n = \sqrt{(-1)^2+(\sqrt{3})^2} = 2$

$t_r = \frac{1.8}{\omega_n} = \frac{1.8}{2} \leq 0.9$

$\theta = \sin^{-1}\zeta$

$\zeta = \sin\theta = \frac{\sigma}{\omega_n} = \frac{1}{2} = 0.5$

$t_s = \frac{4.6}{\sigma} = 4.6$ ✓ (1% settling time)

$M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}} = e^{\frac{-\pi(-0.5)}{\sqrt{1-(-0.5)^2}}} = 0.163$

SEE MATLAB:

$z_{lead} = \omega_n / 6.25 = 2/6.25 = 0.32$

$p_{lead} = 4 * z_{lead} = 2/6.25)(4) = 1.28$

$K = 3.265$

$$D_c(s) = 3.265\frac{s+0.32}{s+1.28}$$

```
% Problem 5.25 b
% Design lead compensator Dc(s) = k * (s+z) / (s+p)
% Requirement: s = -1 +- j * sqrt(3)

clear; clc;
wn = 2;
zeta = 0.5;
s = tf('s');
G = 1/(s*(s+1));
figure(1);

% % Step 2 - Plot uncompensated root locus
rlocus(G);
title('Uncompensated Root Locus');

% % Step 3 - can't meet specs with uncompensated system, so pick starting
% % values for lead compensator

z_lead = wn/6.25; % Rule of thumb: initially 25% of natural frequency.
p_lead = 4*z_lead; % Rule of thumb: initially placing pole at 10*z;
D_lead = (s + z_lead)/(s + p_lead);

figure(2);

% % Step 4 - Examine root locus and see if we can place the poles somewhere
% % where the specs will be met.
rlocus(D_lead*G)

% Step 5 - Select K value
K = 3.2653;


% This portion was generated by AI to graph the point nicely
% Find the closed-loop poles
closed_loop_poles = pole(feedback(K * D_lead * G, 1));

% Plot the marker first
figure(2);
rlocus(D_lead * G);
sgrid(zeta, wn);
hold on;
h = plot(real(closed_loop_poles(1)), imag(closed_loop_poles(1)), 'ro', 'MarkerSize',↙
8, 'LineWidth', 2);

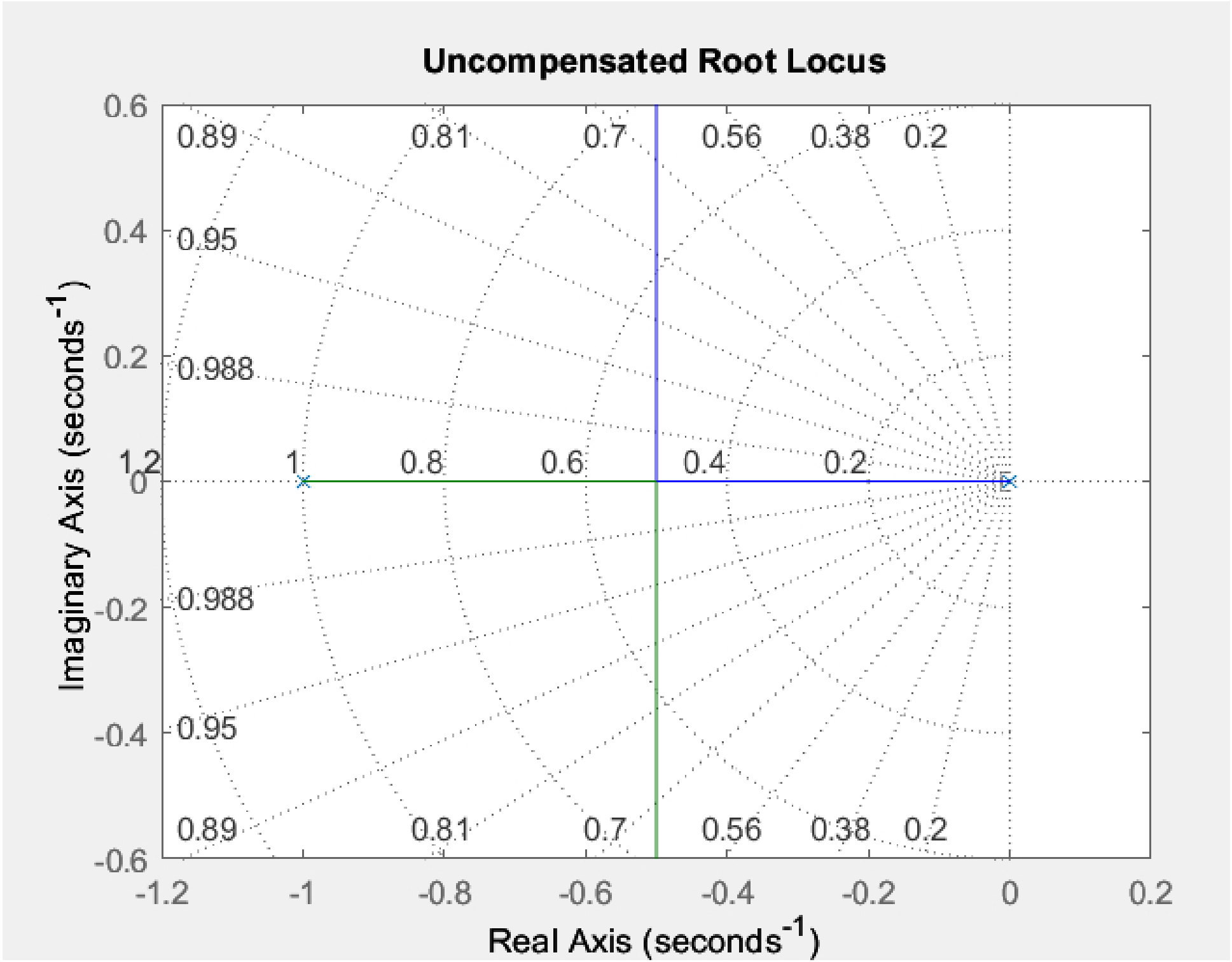
% Automatically generate the text data tip on the plot
row = dataTipTextRow('Gain', repmat(K, size(closed_loop_poles(1))));
row2 = dataTipTextRow('Pole', closed_loop_poles(1));
h.DataTipTemplate.DataTipRows(end+1) = row;
h.DataTipTemplate.DataTipRows(end+1) = row2;
```

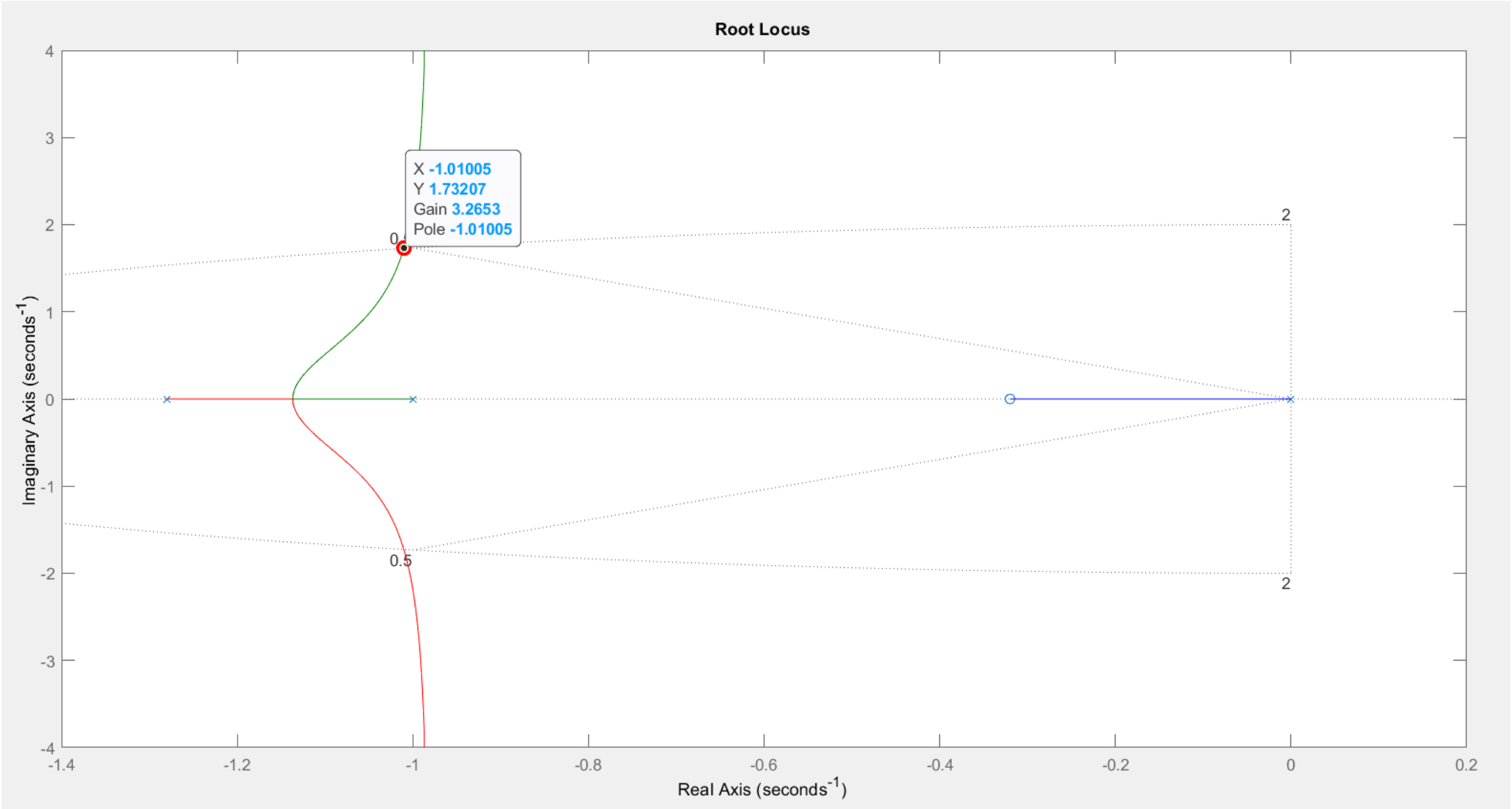
```
% Create the permanent data tip box at that coordinate
dt = datatip(h, real(closed_loop_poles(1)), imag(closed_loop_poles(1)));

hold off;



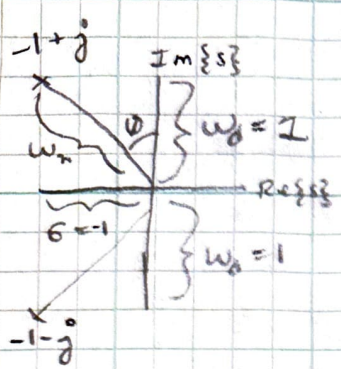

% LAG compensator not asked for
% % Step 6 - Plot step response with full lead compensator. Ensure
% % specifications are met.
% sys = feedback(K*D_lead*G, 1);
% figure(3);
% step(sys)
% % Step 7 - analyze steady state error
% Tau_ten = 50/p_lead;
% t=0:0.001:Tau_ten;
% % Apply ramp input (u(t) = t)
% y=lsim(sys, t, t);
% figure(4);
% plot(t,y,'b',t,t,'r')
% legend("Response", "Ref input")
```

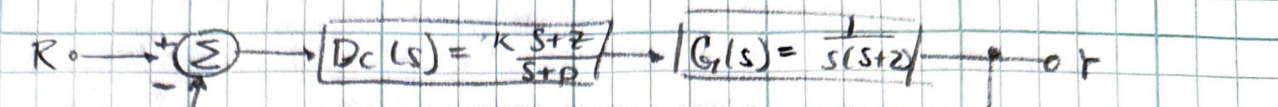

Uncompensated Root Locus
Imaginary Axis (seconds-1)
Real Axis (seconds-1)
0.89
0.81
0.7
0.56
0.38
0.2
0.95
0.988
1
0.8
0.6
0.4
0.2
0
0.6
0.4
0.2
-0.2
-0.4
-0.6
-1.2
-1
-0.8
-0.6
-0.4
-0.2
0


Root Locus
X -1.01005
Y 1.73207
Gain 3.2653
Pole -1.01005
0.5
2
2
0.5
Imaginary Axis (seconds-1)
Real Axis (seconds-1)
4
3
2
1
0
-1
-2
-3
-4
-1.4
-1.2
-1
-0.8
-0.6
-0.4
-0.2
0
0.2

5.27) Assume the closed-loop system of has a feed forward function $G_1(s) = \frac{1}{s(s+2)}$ [4/4]

Design a lag compensation so the dominant poles of the closed-loop system are located at $s = -1 \pm j$ and the steady state error for a unit-ramp input is less than 0.2.





$\omega_n = \sqrt{1^2 + 1^2} = \sqrt{2}$ (6=)

$t_r = \frac{1.8}{\sqrt{2}} = 1.27$

$\zeta = \sin\phi = \frac{1}{\sqrt{2}} = 0.707$

$t_s = \frac{4.6}{6} = 4.6$

$M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}} = 0.043 = 4.3\%$

$e_{ss} < 0.2$

$e_{ss}(t) = \lim_{s\to 0} sE(s)$ $\quad E = \frac{R}{1+D_cG}$

$= \lim_{s\to 0} \frac{s}{1+D_cG_1} R$

$= \lim_{s\to 0} \frac{s}{1 + \frac{k(s+z)}{s(s+p)(s+2)}} R = \lim_{s\to 0} \frac{s}{\frac{s(s+p)(s+2)+k(s+z)}{s(s+p)(s+2)}} R$

$= \lim_{s\to 0} \frac{s^2(s+p)(s+2)}{s(s+p)(s+2)+k(s+z)} \frac{1}{s^2}$

$e_{ss} = \frac{2p}{kz}$

$k = 2$
$z = 0.0071$
$p = 0.0014$

$D_c(s) = 2\frac{s+0.0071}{p+0.0014}$; $e_{ss} = 0.1994$

* See MATLAB for derivation

```
% Problem 5.27
% Design lead compensator Dc(s) = k * (s+z) / (s+p), G(s) = 1/(s(s+2))
% Requirement: s = -1 +- j
% ess < 0.2

clear; clc;
wn = sqrt(2);
zeta = 1/sqrt(2); % 0.707
s = tf('s');
G = 1/(s*(s+2));
figure(1);

% % Step 2 - Plot uncompensated root locus
rlocus(G);
title('Uncompensated Root Locus');

% % Step 3 - can't meet specs with uncompensated system, so pick starting
% % values for lead compensator

% z_lead = wn/4; % Rule of thumb: initially 25% of natural frequency.
% p_lead = 10*z_lead; % Rule of thumb: initially placing pole at 10*z;
% D_lead = (s + z_lead)/(s + p_lead);

z_lag = wn / 200;
p_lag = z_lag / 5;
D_lag = (s+z_lag)/(s+p_lag);

figure(2);

rlocus(D_lag * G)




% Define the target pole location
s_target = -1 + 1j;

% Compute the root locus for a fine vector of gains
k_vec = linspace(0, 20, 50000);
[r, k] = rlocus(D_lag * G, k_vec);

% Find the point on the root locus closest to s_target
[min_dist, idx_linear] = min(abs(r(:) - s_target));
[row, col] = ind2sub(size(r), idx_linear);

K_exact = k(col);
achieved_pole = r(row, col);
```

```
% Display the results
fprintf('\n--- Root Locus Verification ---\n');
fprintf('Target Pole:    %.4f + %.4fi\n', real(s_target), imag(s_target));
fprintf('Achieved Pole:  %.4f + %.4fi\n', real(achieved_pole), imag(achieved_pole));
fprintf('Required Gain:  K = %.4f\n', K_exact);
fprintf('Distance Error: %.6f\n', min_dist);

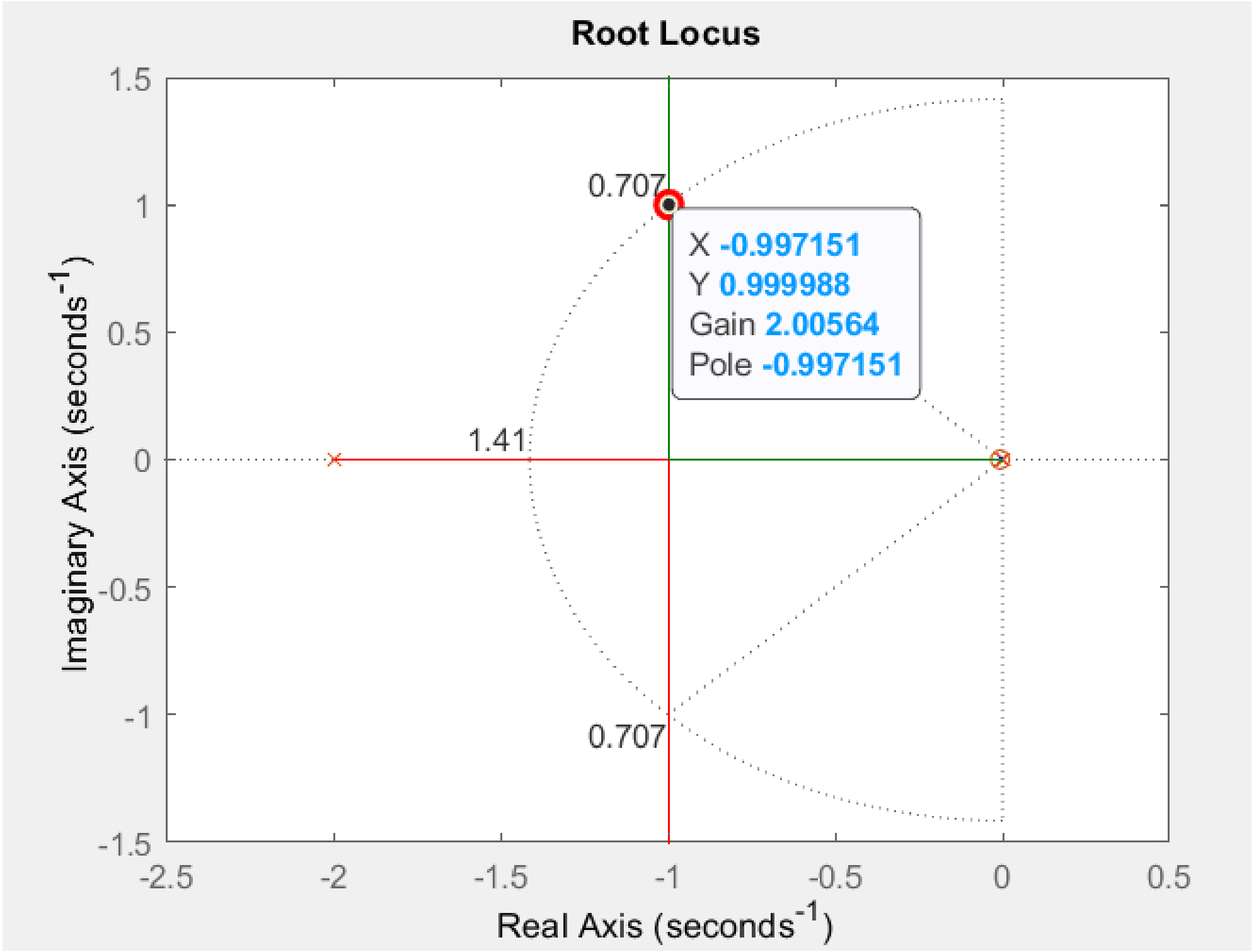
if min_dist < 0.05
    fprintf('SUCCESS: The root locus passes through s = -1 + j!\n');
else
    fprintf('WARNING: The root locus misses s = -1 + j (Error too large).\n');
end

% % Step 5 - Select K value
K = K_exact;


% Step 6 - Plot step response with full lead compensator. Ensure
% specifications are met.
% sys = feedback(K*D_lag*G, 1);
% figure(3);
% step(sys)
%
% % Step 7 - analyze steady state error
% Tau_ten = 50/p_lag;
% t=0:0.001:Tau_ten;
% % Apply ramp input (u(t) = t)
% y=lsim(sys, t, t);
% figure(4);
% plot(t,y,'b',t,t,'r')
% legend("Response", "Ref input")

ess = (2 * p_lag) / (K * z_lag);
fprintf('ess = %.4f\n', ess)
```

```
--- Root Locus Verification ---
Target Pole:     -1.0000 + 1.0000i
Achieved Pole:   -0.9972 + 1.0000i
Required Gain:   K = 2.0056
Distance Error: 0.002849
SUCCESS: The root locus passes through s = -1 + j!
ess = 0.1994
```


Root Locus
1.5
1
0.5
0
-0.5
-1
-1.5
-2.5
-2
-1.5
-1
-0.5
0
0.5
Imaginary Axis (seconds-1)
Real Axis (seconds-1)
0.707
1.41
0.707
X -0.997151
Y 0.999988
Gain 2.00564
Pole -0.997151